D02 · Supply Planning & Scheduling

03 · Production scheduling & sequence optimization

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1 WHAT IT IS

Production scheduling and sequence optimization order the jobs on each resource to minimize sequence-dependent setup and changeover time while meeting due dates — treating the sequence itself as a decision variable — instead of running jobs first-in-first-out by order date and absorbing whatever changeover the arbitrary order happens to cost.

2 WHY IT MATTERS

Johnson 1954 **(Johnson, 1954 — peer-reviewed)** founded production scheduling by proving that the sequence of jobs through stages is itself optimizable and that the right order materially reduces total completion time — the sequence is a lever, not a given. When changeovers are sequence-dependent (a common case: color, grade, temperature, or tooling transitions cost different amounts depending on what ran before), FIFO scheduling leaves large amounts of avoidable setup time on the floor. That lost time is lost capacity: **changeover that could be 8% of available hours under a good sequence can balloon to 20%+ under FIFO**, and the plant pays for capacity it sets up and tears down instead of producing.

3 HOW IT IS DONE

1. **Build the sequence-dependent setup matrix.** Quantify the changeover time and cost for each transition from product A to product B on each resource. Without this matrix the scheduler is blind to the very cost it should be minimizing — the matrix is the prerequisite, not a refinement.
2. **Treat the sequence as the decision variable.** Optimize the order of jobs to minimize total setup subject to due dates, not the other way around. FIFO and earliest-due-date are heuristics that ignore changeover entirely; the sequence is where the capacity is won or lost.
3. **Balance setup minimization against due-date adherence.** The shortest-setup sequence may miss promises; the objective is the best total of changeover plus tardiness, not setup alone. A schedule that saves changeover by being chronically late has optimized the wrong thing.
4. **Identify and protect the bottleneck.** Sequence the constraint resource first and subordinate the rest to it. An hour of changeover saved at the bottleneck is an hour of plant capacity gained; an hour saved at a non-bottleneck is free time that changes nothing.
5. **Use campaign and wheel logic where transitions are costly.** Group like products into runs or a fixed production wheel to amortize expensive changeovers, accepting some inventory in exchange for reclaimed capacity. Size the campaign against the changeover-to-holding trade-off, not by habit.
6. **Re-optimize on disruption, not on a fixed timer.** When a rush order, breakdown, or material delay hits, re-sequence the affected horizon rather than forcing the old sequence. A scheduler that cannot re-optimize quickly defaults back to FIFO under pressure — exactly when sequence matters most.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A process plant scheduled a multi-grade line FIFO by order date — **baseline: changeover consuming 19% of available hours, frequent grade thrashing, due-date attainment 86%**. It built the setup matrix, sequenced the bottleneck to minimize transitions subject to due dates, and adopted campaign logic for the costliest changeovers. Results: **changeover 19% -> 9% of available hours, effective capacity up ~12% with no capital, and due-date attainment 86% -> 93%** — capacity reclaimed from the sequence, not bought.

IN PRACTICE

Maintain the setup matrix

Keep the sequence-dependent changeover times current as products, tooling, and processes change. A scheduler optimizing against a stale matrix produces a confident, wrong sequence — the matrix is master data and decays like all master data.

Multi-objective scheduling

Optimize setup, due-date adherence, and WIP together rather than any one alone. The right schedule is the best total, and weighting the objectives is a business decision — a pure setup-minimizer that misses promises is as broken as a pure due-date sort that thrashes changeover.

Schedule to the constraint

Identify the bottleneck and build the schedule around its optimal sequence, subordinating upstream and downstream resources to feed and follow it. Capacity gains come from the constraint; effort spent optimizing non-constraints is motion without movement.

Fast re-sequencing on disruption

Build the capability to re-optimize the affected horizon in minutes when reality diverges. A scheduler that takes hours to rebuild gets bypassed in a disruption, and the floor reverts to FIFO precisely when an optimized sequence would matter most.

WHAT YOU WOULD USE

PRODUCTION SCHEDULING / APS

Blue Yonder / SAP PP-DS / o9 — enterprise tier-1 — Finite-capacity scheduling with sequence-dependent setup optimization, bottleneck modeling, and fast re-optimization. **(Blue Yonder, SAP, o9 — vendor)** The choice when scheduling must integrate with the broader supply plan and execution.

Siemens Opcenter / dedicated APS-scheduling — Strong detailed-scheduling engines with rich setup-matrix and shop-floor modeling. **(Siemens, dedicated APS vendors — vendor)** A strong fit where the scheduling complexity (sequence-dependent setups, complex routings) is the core problem.

Scheduling-optimization / solver specialists — Specialist scheduling and optimization-solver tools for highly constrained or custom sequencing problems. **(specialized vendors — vendor)** The vertical option when the sequencing problem is unusual enough that packaged APS heuristics fall short.

For operations that don't need a platform — A setup matrix plus campaign/wheel logic applied by a planner in a scheduling spreadsheet, focused on the bottleneck. Even a manual sequence built from a real setup matrix beats FIFO decisively; the matrix and the bottleneck focus, not the solver, deliver the first and largest gain.

4 SO WHAT

Operations that optimize the sequence reclaim the capacity FIFO spends on avoidable changeover; operations that schedule by arrival order pay for setup time they could have eliminated. Sequence-dependent setup is the most overlooked source of free capacity on the floor — no capital, no headcount, just the order of the jobs. The plant that schedules FIFO is buying capacity it already owns and throws away every shift.

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